

The conjectured P-recursive recurrence for OEIS A026110, proved

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Abstract

OEIS A026110 carries a conjectured recurrence of order 4 whose coefficients are polynomials in n rather than constants. It is true, and it is decidable rather than empirical. The entry states its generating function as a fact, and that function is algebraic; the conjecture then becomes the vanishing of an explicit element of an algebraic function field, which is settled symbolically. The result is an identity, not a run of verified terms: it holds at every index at once.

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1 The conjecture

OEIS A026110 is “ $a(n)$ = number of $(s(0), s(1), \dots, s(n))$ such that every $s(i)$ is a nonnegative integer, $s(0) = 0$, $s(1) = 1$, $s(n) = 4$, $|s(i) - s(i-1)| \leq 1$ for $i \geq 2$. Also $a(n) = T(n, n-4)$, where T is the array defined in A026105.”. It has offset 4 and begins

$$1, 4, 15, 50, 160, 496, 1509, 4530, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

$$\begin{aligned} \text{Conjecture: } & -(n+6) \cdot (n-4) \cdot a(n) + (4n^2 - 51n) \cdot a(n-1) + (-2n^2 + 11n + 18) \cdot a(n-2) \\ & - (4n-1) \cdot (n-3) \cdot a(n-3) + 3 \cdot (n-3) \cdot (n-4) \cdot a(n-4) = 0. \quad \text{— R. J. Mathar, Jun 23 2013} \end{aligned}$$

As of the “Last modified” line on the live entry (Aug 27 2022 03:11:12, revision 14) this is still recorded as a conjecture, and nothing on the entry records it as proved.

2 The generating function

The entry gives, as a statement of fact and not as a conjecture,

$$A(x) = \frac{(1-x) \left(-x - \sqrt{-3x^2 - 2x + 1} + 1 \right)^5}{32x^6}.$$

This is an algebraic function. Before it is used for anything it is checked against the entry’s own data: its Taylor coefficients reproduce every one of the 26 published terms exactly, in exact arithmetic. That check is not a formality — on A116388 the generating function an entry states as fact does NOT generate the terms it publishes, and a proof built on it would have been a proof from a false premise.

3 A P-recursive claim is the vanishing of one function

Lemma 1. *Let $\theta = x \frac{d}{dx}$, so θ acts on x^m as multiplication by m . For a polynomial p and an integer $i \geq 0$,*

$$\sum_n p(n) a(n-i) x^n = x^i (p(\theta + i)A)(x).$$

Proof. Write $m = n - i$. Then $\sum_m p(m+i)a(m)x^{m+i} = x^i \sum_m p(m+i)a(m)x^m$, and $p(\theta + i)$ applied to $A = \sum_m a(m)x^m$ multiplies the m -th coefficient by $p(m+i)$. $\square$

Put $b(n) = \sum_{i=0}^4 p_i(n) a(n-i)$, the left-hand side of the conjecture, and $B(x) = \sum_n b(n)x^n$. By Lemma 1,

$$B(x) = \sum_{i=0}^4 x^i (p_i(\theta + i)A)(x),$$

an explicit element of the algebraic function field generated by A . The conjecture asserts $b(n) = 0$; so it holds for all n exactly when $B = 0$, and for all $n > d$ exactly when B is a polynomial of degree d . Both are decided symbolically, with no truncation and no numerics.

4 The theorem

Theorem 2.

$(-n^2 - 2n + 24) a(n) + (4n^2 - n - 51) a(n-1) + (-2n^2 + 11n + 18) a(n-2) + (-4n^2 + 13n - 3) a(n-3) + (3n^2 -$
for every $n \geq 4$.

Proof. B was formed as above and simplified in the function field. Here B is identically zero, so the recurrence holds for EVERY n at which its indices are defined. Since A reproduces the entry's published terms, b is the sequence the entry's conjecture is about, and the statement follows. $\square$

5 Verification

First, the generating function reproduces all 26 published terms exactly. Second, the conjectured recurrence was evaluated directly on those terms, with no generating function involved, and holds wherever the proved range and the data overlap. Third, the residual B was computed symbolically rather than from a truncated series, and its series expansion was taken independently to confirm the coefficients it predicts.

References

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- [3] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011.